FIITJEE Students From Classroom / Integrated School Programs have secured to 2 Zonal, 6 State & 18 City Topper Ranks. 33 in Top 100, 78 in Top 200 and 205 in Top 500
All India Ranks bagged by FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2016

ANSWERS, HINTS & SOLUTIONS

FULL TEST–II
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	D	D	B
2.	B	C	D
3.	C	C	B
4.	B	C	A
5.	B	D	B
6.	A	D	B
7.	A, B, C	C	A, C, D
8.	A, B, C, D	B, C	A, B, C
9.	B, C	A	C, D
10.	A	D	A
11.	C	C	B
12.	B	B	A
13.	B	B	D
14.	C	A	D
1.	(A) → p, q, r, s, (B) → p, q, r, s, (C) → p, q, r, s, (D) → r	(A) → p, q, r, s, t, (B) → p, r, (C) → r, t, (D) → p, q, s, t	(A) → q, (B) → s, (C) → t, (D) → p
2.	(A) → q, r, (B) → p, r, (C) → p, r, (D) → s	(A) → q, (B) → r, (C) → p, t, (D) → s, t	(A) → r, (B) → s, (C) → p, (D) → q
1.	7	0	9
2.	0	8	2
3.	4	4	4
4.	1	4	6
5.	1	5	3

Chemistry

PART – II

SECTION – A

1. D
It is a chain silicate.

2. C
Total $\left[H^+\right] = C_{H^+ due\ to\ HCl} + \left[OH^-\right]$.

3. C
(A) Both carbonates give oxides and CO_2 gas.
(B) Both bicarbonates give carbonates, CO_2 and H_2O
(C) $MgCl_2.6H_2O \xrightarrow{\Delta} MgCl_2.H_2O + 5H_2O$

$\rightarrow Mg(OH)Cl + HCl$

$Na_2SO_4.10\,H_2O \xrightarrow{\Delta} Na_2SO_4 + 10\,H_2O$

(D) Both the sulphates give oxides and SO_3

4. C
$A \rightarrow 3B$

$a \quad 0$

$a-x \quad 3x$

$V_0 \alpha (a)$ and $V_t \alpha (a+5x)$

$\therefore$ equation is $K = \frac{2.303}{t}\log\frac{5V_0}{6V_0 - V_t}$

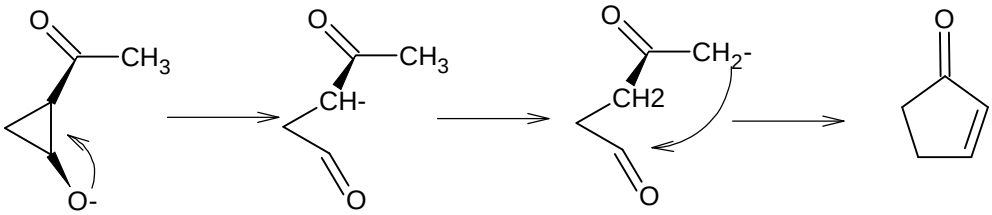
Substituting $V_t = 2V_0$ at $t = 971\text{sec}$

$K = \frac{2.303}{971} \times \log\frac{5}{4} = \frac{2.303}{971} \times .0971 = 2.303 \times 10^{-4}$

$\therefore t_{1/2} = \frac{.693}{2.303 \times 10^{-4}} = 3010\,\text{sec}$

5. D
Ortho can be easily reduced than para isomer, more rings can also be reduced easily

6. D



7. C
$NH_3 \uparrow \rightleftharpoons \frac{1}{2}N_2 \uparrow + \frac{3}{2}H_2 \uparrow$

$n \quad 0 \quad 0$

$n(1-\alpha) \quad \frac{n\alpha}{2} \quad \frac{3n\alpha}{2}$

Partial pressure at equation $\left(\frac{1-\alpha}{1+\alpha}\right)P; \frac{\alpha}{2(1+\alpha)}P; \frac{3}{2}\left(\frac{\alpha}{1+\alpha}\right)P$

$$K_p = \frac{\left(P_{N_2}\right)^{1/2}\left(P_{H_2}\right)^{3/2}}{P_{NH_3}} = \frac{3\sqrt{3}P}{4}\left[\frac{\alpha^2}{1-\alpha^2}\right]$$

$$\therefore \alpha = \left[1+\frac{3\sqrt{3}}{4}\frac{P}{K_p}\right]^{-1/2}$$

8. B, C
Polar forms are more stable in polar solvents.

9. A
Volume of H_2SO_4 used at equivalence point is 15mL.
$15\times 0.2\times 2 = 20\times conc.$
Conc. = 0.3

SECTION – B

2. (A) → q, (B) → r, (C) → p, t, (D) → s, t
(p) $2H^+ + 2e^- \rightarrow H_2$

$$E = -\frac{0.0591}{2}\log\frac{1}{\left[10^{-7}\right]^2} = -0.414\ V$$

(q) $H_2 \longrightarrow 2H^+ + 2e^-$
[It will be oxidation potential]
(r) $2Hg + Cl_2 \longrightarrow Hg_2^{2+} + 2Cl^-$
(s) $Ag^+ + e^- \longrightarrow Ag$

$$\left(Ag^+\right) = \sqrt{10^{-16}} = 10^{-8}\ M$$

SECTION – C

4. The coordination compound is $Pd\left[PdF_6\right]$. The coordination number of $Pd\left(IV\right)$ is six:

Mathematics

PART – III

SECTION – A

1. **Use expansion of** $\sin^{-1}x$ & $\tan^{-1}x$ to get the result

2. Plot the curve $y-1=\frac{1}{x}$ and translate the axis to get required area.

3. $2x^3y\,dy+3x^2y^2dx=\frac{2yxdx-x^2dy}{y^2}$

$\int d\left(x^3y^2\right)=\int d\left(\frac{x^2}{\hat{j}}\right)$

4. $\frac{1-|x|}{2-|x|}\geq 0 \Rightarrow \frac{|x|-1}{|x|-2}\geq 0$

$\Rightarrow use\ wavy\ curve\ of\ \frac{(x-1)(x+1)}{(x-2)(x+2)}\geq 0$

(10- 11)

$Use\ a=k-3e$

$b=k-e$

$c=k+e \qquad in\ \ d=a^2+b^2+c^2$

$d=k+3e$

(12 – 14)

Plot the graphs using periodicity

SECTION – B

Matrix Match – 2

Use $(2a)^2+(3b)^2+12ab-c^2=0$

$\Rightarrow (2a+3b-c)(2a+3b+c)=0$

SECTION – C

[Integer Type]

2. Use director circle concept.

3. Use $-\sqrt{a^2+b^2}\leq a\sin\theta+b\cos\theta\leq\sqrt{a^2+b^2}$